

Finite Element Method

FEM for Multidimensional Vector Field Problems – Part 2
MECH4103 — Spring 2026

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- The triangular three-node element is a **linear displacement element**; the strains are constant in the element. Nodes must be numbered **counterclockwise**.
- Each node has two degrees of freedom, so $\mathbf{d}^e$ consists of six terms:

$$\mathbf{d}^e = [u_{x1}^e, u_{y1}^e, u_{x2}^e, u_{y2}^e, u_{x3}^e, u_{y3}^e]^T. \quad (1)$$

- The displacement field in the element is

$$\begin{bmatrix} u_x \\ u_y \end{bmatrix}^e = \begin{bmatrix} N_1^e & 0 & N_2^e & 0 & N_3^e & 0 \\ 0 & N_1^e & 0 & N_2^e & 0 & N_3^e \end{bmatrix} \mathbf{d}^e.$$

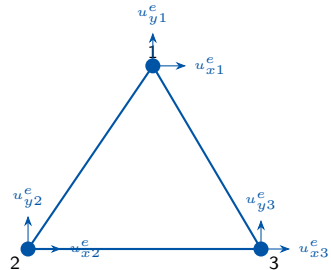


Figure 9.7 A single triangular finite element.

- Applying the symmetric gradient operator gives the strain–displacement relation:

$$\begin{bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \gamma_{xy} \end{bmatrix}^e = \begin{bmatrix} N_{1,x}^e & 0 & N_{2,x}^e & 0 & N_{3,x}^e & 0 \\ 0 & N_{1,y}^e & 0 & N_{2,y}^e & 0 & N_{3,y}^e \\ N_{1,y}^e & N_{1,x}^e & N_{2,y}^e & N_{2,x}^e & N_{3,y}^e & N_{3,x}^e \end{bmatrix} \mathbf{d}^e, \quad (2)$$

where $N_{I,x}^e = \partial N_I^e / \partial x$ and $N_{I,y}^e = \partial N_I^e / \partial y$. Evaluating with $2A = \det \begin{bmatrix} 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \\ 1 & x_3 & y_3 \end{bmatrix}$ yields

$$\boldsymbol{\varepsilon}^e = \frac{1}{2A^e} \begin{bmatrix} y_{23}^e & 0 & y_{31}^e & 0 & y_{12}^e & 0 \\ 0 & x_{32}^e & 0 & x_{13}^e & 0 & x_{21}^e \\ x_{32}^e & y_{23}^e & x_{13}^e & y_{31}^e & x_{21}^e & y_{12}^e \end{bmatrix} \mathbf{d}^e, \quad (3)$$

where $x_{IJ}^e = x_I^e - x_J^e$. This defines the $\mathbf{B}^e$ matrix. Note that $\mathbf{B}^e$ is not a function of x or y — the strain is constant in the element.

- The stiffness matrix is given by $\mathbf{K}^e = \int_{\Omega^e} (\mathbf{B}^e)^T \mathbf{D}^e \mathbf{B}^e d\Omega$. For a low-order element with constant material properties and unit thickness:

$$\mathbf{K}^e = A^e (\mathbf{B}^e)^T \mathbf{D}^e \mathbf{B}^e.$$

The stiffness matrix is 6×6 and is typically evaluated by computer.

- The element body force matrix is given by

$$\mathbf{f}_{\Omega}^e = \int_{\Omega} (\mathbf{N}^e)^T \mathbf{b} \, d\Omega. \quad (4)$$

There are two ways of evaluating this matrix:

- ❶ by direct numerical integration, and
 - ❷ by interpolating $\mathbf{b}$, usually with a linear function, and integrating the result in closed form. In direct integration, interpolation is still often required as the body forces may only be given at discrete points.
- Evaluation in closed form is most conveniently done using triangular coordinates. We interpolate the body force in the element by the linear shape functions as

$$\mathbf{b} = \begin{bmatrix} b_x \\ b_y \end{bmatrix} = \sum_{I=1}^3 N_I^{3T} \begin{bmatrix} b_{xI} \\ b_{yI} \end{bmatrix}, \quad (5)$$

where b_{xI} and b_{yI} are the x - and y -components of the body force at node I .

Substituting (5) into (4) yields

$$\mathbf{f}_{\Omega}^e = \int_{\Omega^e} \begin{bmatrix} N_1^{3T} & 0 \\ 0 & N_1^{3T} \\ N_2^{3T} & 0 \\ 0 & N_2^{3T} \\ N_3^{3T} & 0 \\ 0 & N_3^{3T} \end{bmatrix} \sum_{I=1}^3 N_I^{3T} \begin{bmatrix} b_{xI} \\ b_{yI} \end{bmatrix} d\Omega = \frac{A^e}{12} \begin{bmatrix} 2b_{x1} + b_{x2} + b_{x3} \\ 2b_{y1} + b_{y2} + b_{y3} \\ b_{x1} + 2b_{x2} + b_{x3} \\ b_{y1} + 2b_{y2} + b_{y3} \\ b_{x1} + b_{x2} + 2b_{x3} \\ b_{y1} + b_{y2} + 2b_{y3} \end{bmatrix}. \quad (6)$$

- The boundary force matrix is given by

$$\mathbf{f}_{\Gamma}^e = \int_{\Gamma_t^e} (\mathbf{N}^e)^T \bar{\mathbf{t}} d\Gamma. \quad (7)$$

- Consider the triangular element with traction applied to the edge joining nodes 1 and 2. The Kronecker delta property ensures N_3^e vanishes along this edge. The shape functions on the edge reduce to $N_1^{2L} = 1 - \xi$ and $N_2^{2L} = \xi$.
- Integral (7) becomes (with $d\Gamma = l d\xi$, l = edge length):

$$\mathbf{f}_{\Gamma}^e = \frac{l}{6} \begin{bmatrix} 2t_{x1} + t_{x2} \\ 2t_{y1} + t_{y2} \\ t_{x1} + 2t_{x2} \\ t_{y1} + 2t_{y2} \\ 0 \\ 0 \end{bmatrix}.$$

For a constant traction $t_{x1} = t_{x2} = \bar{t}_x$, $t_{y1} = t_{y2} = \bar{t}_y$:

$$\mathbf{f}_{\Gamma}^e = \frac{l}{6} \begin{bmatrix} \bar{t}_x \\ \bar{t}_y \\ \bar{t}_x \\ \bar{t}_y \\ 0 \\ 0 \end{bmatrix}$$

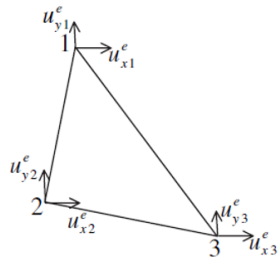
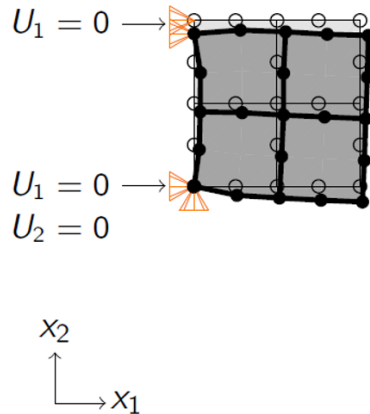


Figure 9.7 A single triangular finite element.

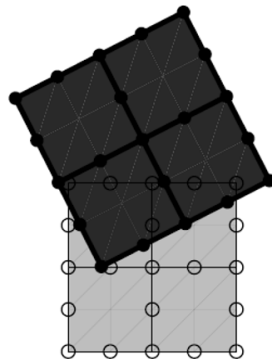
Figure 9.8 Triangular three-node element showing nodal displacements and nodal forces (shown collinear, though they are usually not).

- For a **2D solid**, 3 rigid body modes exist (2 translations + 1 rotation). With 3 displacement BCs, all rigid body motions are eliminated and the stiffness matrix becomes regular.
- For a **3D solid**, there are 6 rigid body modes: 3 translations and 3 rotations.



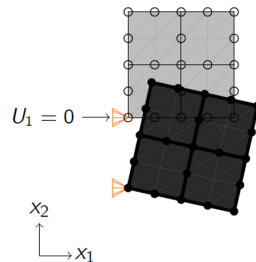
Example: 3 displacement BCs suppress all 3 rigid body modes in 2D.

- For a 2D solid, three rigid body modes exist.
- A 2D solid *without* any displacement BCs can translate horizontally, vertically, and rotate.
- A linear static analysis will not produce a result — the stiffness matrix is singular.

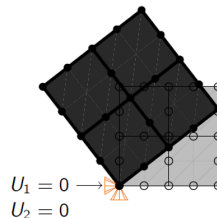


No displacement BCs: all three rigid body modes are active.

- The model in light gray is the undeformed configuration; the dark gray result comes from a natural frequency analysis used to identify rigid body modes.
- One displacement BC ($U_1 = 0$) eliminates one rigid body mode — the solid can no longer translate freely in the x_1 -direction.
- When the lower left corner is fully fixed ($U_1 = U_2 = 0$), only the rotational rigid body mode remains. Rotation about the lower left corner is still arbitrary, so the solution is not unique.



One BC: translation suppressed, rotation remains.



- Consider the plate with a hole with loads applied at the top and bottom.
- Sides AD and BC are traction free; no action is needed to enforce a homogeneous natural BC.
- Sides CD and AB are also natural boundaries, but the tractions must be incorporated through the boundary force matrix.
- These BCs alone do not suppress rigid body motion — the stiffness matrix is **singular**.
- To eliminate rigid body motion, at least **three** nodal displacement components must be specified. One valid choice is

$$u_{xA} = u_{yA} = u_{yB} = 0.$$

Replacing $u_{yB} = 0$ by $u_{xB} = 0$ leaves $\mathbf{K}$ singular because rotation is not prevented.

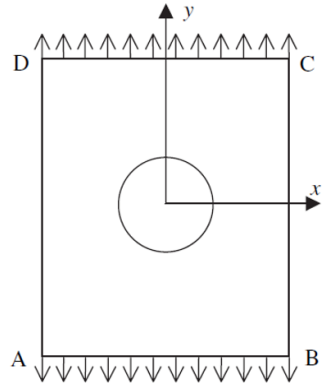
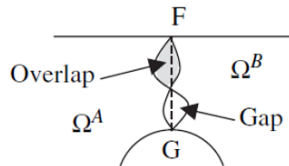
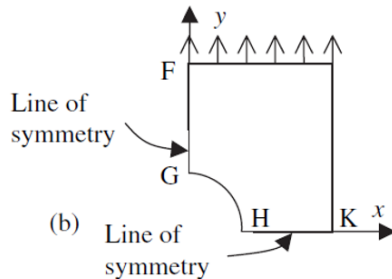


Plate with hole: loads on top (C) and bottom (A/B).

- Another approach is to exploit symmetry. The problem has two lines of symmetry: FG and HK — see figure (b).
- Along a line of symmetry, the displacement component **normal** to the line must vanish. A nonzero normal displacement would create gaps or overlaps in the symmetric subdomain, violating compatibility — see figure (c).
- The symmetry boundary conditions are:

$$u_x = 0 \quad \text{and} \quad t_y = \sigma_{xy} = 0 \quad \text{on FG,}$$

$$u_y = 0 \quad \text{and} \quad t_x = \sigma_{xy} = 0 \quad \text{on HK.}$$

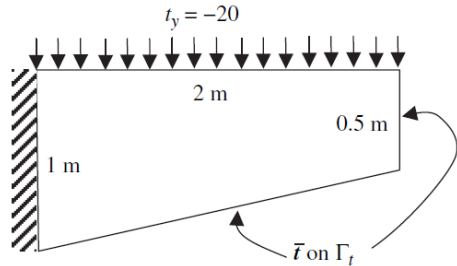


Consider a linear elasticity problem on the trapezoidal panel in Figure 9.11:

- Vertical left edge: fixed.
- Bottom and right vertical edges: traction free ($\bar{\mathbf{t}} = \mathbf{0}$).
- Top edge: traction $\bar{t}_y = -20 \text{ N m}^{-1}$.
- Material: $E = 3 \times 10^7 \text{ Pa}$, $\nu = 0.3$, plane stress.
- Discretised with one quadrilateral element.

The constitutive matrix is

$$\mathbf{D} = \frac{E}{1 - \nu^2} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{bmatrix} = 3.3 \times 10^7 \begin{bmatrix} 1 & 0.3 & 0 \\ 0.3 & 1 & 0 \\ 0 & 0 & 0.35 \end{bmatrix}.$$



Trapezoidal panel (Figure 9.11): fixed left edge, traction on top.

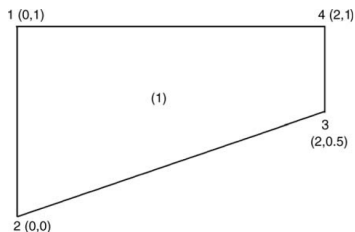


Figure 9.12 Finite element mesh for Example 9.2.

Figure 9.12 FE mesh for Example 9.2. Nodal coordinates (m): 1(0,1), 2(0,0), 3(2,0.5), 4(2,1).

The coordinate matrix is

$$\begin{bmatrix} \mathbf{x}^e & \mathbf{y}^e \end{bmatrix} = \begin{bmatrix} x_1^e & y_1^e \\ x_2^e & y_2^e \\ x_3^e & y_3^e \\ x_4^e & y_4^e \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \\ 2 & 0.5 \\ 2 & 1 \end{bmatrix}.$$

The Lagrangian shape functions in the parent element are

$$N_1^{4Q}(\xi, \eta) = \frac{1}{4}(1 - \xi)(1 - \eta),$$

$$N_2^{4Q}(\xi, \eta) = \frac{1}{4}(1 + \xi)(1 - \eta),$$

$$N_3^{4Q}(\xi, \eta) = \frac{1}{4}(1 + \xi)(1 + \eta),$$

$$N_4^{4Q}(\xi, \eta) = \frac{1}{4}(1 - \xi)(1 + \eta).$$

The Jacobian matrix is given by

$$\mathbf{J}^e = \begin{bmatrix} \frac{\partial N_1^{4Q}}{\partial \xi} & \frac{\partial N_2^{4Q}}{\partial \xi} & \frac{\partial N_3^{4Q}}{\partial \xi} & \frac{\partial N_4^{4Q}}{\partial \xi} \\ \frac{\partial N_1^{4Q}}{\partial \eta} & \frac{\partial N_2^{4Q}}{\partial \eta} & \frac{\partial N_3^{4Q}}{\partial \eta} & \frac{\partial N_4^{4Q}}{\partial \eta} \end{bmatrix} \begin{bmatrix} x_1^e & y_1^e \\ x_2^e & y_2^e \\ x_3^e & y_3^e \\ x_4^e & y_4^e \end{bmatrix}$$

$$= \frac{1}{4} \begin{bmatrix} \eta - 1 & 1 - \eta & 1 + \eta & -\eta - 1 \\ \xi - 1 & -\xi - 1 & 1 + \xi & 1 - \xi \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \\ 2 & 0.5 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0.125\eta - 0.375 \\ 1 & 0.125\xi + 0.125 \end{bmatrix}.$$

The determinant and inverse of the Jacobian matrix are

$$|\mathbf{J}^e| = -0.125\eta + 0.375,$$

$$(\mathbf{J}^e)^{-1} = \begin{bmatrix} \frac{1+\xi}{3-\eta} & 1 \\ \frac{8}{\eta-3} & 0 \end{bmatrix}.$$

The strain–displacement matrix in the physical domain is $\mathbf{G}_x = \begin{bmatrix} \partial N/\partial x \\ \partial N/\partial y \end{bmatrix} = \mathbf{J}^{-1} \mathbf{G}_\xi$, which in the parent domain reads

$$\mathbf{B}(\xi, \eta) = \begin{bmatrix} \frac{\partial N_1}{\partial \xi} & 0 & \frac{\partial N_2}{\partial \xi} & 0 & \frac{\partial N_3}{\partial \xi} & 0 & \frac{\partial N_4}{\partial \xi} & 0 \\ 0 & \frac{\partial N_1}{\partial \eta} & 0 & \frac{\partial N_2}{\partial \eta} & 0 & \frac{\partial N_3}{\partial \eta} & 0 & \frac{\partial N_4}{\partial \eta} \\ \frac{\partial N_1}{\partial \eta} & \frac{\partial N_1}{\partial \xi} & \frac{\partial N_2}{\partial \eta} & \frac{\partial N_2}{\partial \xi} & \frac{\partial N_3}{\partial \eta} & \frac{\partial N_3}{\partial \xi} & \frac{\partial N_4}{\partial \eta} & \frac{\partial N_4}{\partial \xi} \end{bmatrix}.$$

The element matrices will be integrated using 2×2 Gauss quadrature:

$$\xi_1 = -\frac{1}{\sqrt{3}}, \quad \xi_2 = \frac{1}{\sqrt{3}}, \quad \eta_1 = -\frac{1}{\sqrt{3}}, \quad \eta_2 = \frac{1}{\sqrt{3}}, \quad W_1 = W_2 = 1.$$

The stiffness matrix is

$$\mathbf{K} = \mathbf{K}^{(1)} = \int_{\Omega} (\mathbf{B}^e)^T \mathbf{D}^e \mathbf{B}^e d\Omega = \int_{-1}^1 \int_{-1}^1 (\mathbf{B}^e)^T \mathbf{D}^e \mathbf{B}^e |\mathbf{J}^e| d\xi d\eta = \sum_{i=1}^2 \sum_{j=1}^2 W_i W_j |\mathbf{J}^e(\xi_i, \eta_j)| (\mathbf{B}^e)^T(\xi_i, \eta_j) \mathbf{D}^e \mathbf{B}^e(\xi_i, \eta_j).$$

At the Gauss point $(\xi_1, \eta_1) = (-1/\sqrt{3}, -1/\sqrt{3})$, the shape function derivatives w.r.t. physical coordinates are

$$\begin{bmatrix} \partial N_1^{4Q}/\partial x & \partial N_2^{4Q}/\partial x & \partial N_3^{4Q}/\partial x & \partial N_4^{4Q}/\partial x \\ \partial N_1^{4Q}/\partial y & \partial N_2^{4Q}/\partial y & \partial N_3^{4Q}/\partial y & \partial N_4^{4Q}/\partial y \end{bmatrix}_{(\xi_1, \eta_1)} = \begin{bmatrix} -0.44 & -0.06 & 0.12 & 0.38 \\ 0.88 & -0.88 & -0.24 & 0.24 \end{bmatrix}.$$

Thus, the strain–displacement matrix at (ξ_1, η_1) is

$$\mathbf{B}^e(\xi_1, \eta_1) = \begin{bmatrix} -0.44 & 0 & -0.06 & 0 & 0.12 & 0 & 0.38 & 0 \\ 0 & 0.88 & 0 & -0.88 & 0 & -0.24 & 0 & 0.24 \\ 0.88 & -0.44 & -0.88 & -0.06 & -0.24 & 0.12 & 0.24 & 0.38 \end{bmatrix}.$$

The stiffness matrix contribution from Gauss point (ξ_1, η_1) is

$$\mathbf{K}^e(\xi_1, \eta_1) = W_1 W_1 (\mathbf{B}^e)^T(\xi_1, \eta_1) \mathbf{D}^e \mathbf{B}^e(\xi_1, \eta_1) |\mathbf{J}^e(\xi_1, \eta_1)|.$$

Repeating for the remaining three Gauss points (ξ_1, η_2) , (ξ_2, η_1) and (ξ_2, η_2) and summing:

$$\mathbf{K}^e = \sum_{i=1}^2 \sum_{j=1}^2 \mathbf{K}^e(\xi_i, \eta_j) = 10^7 \begin{bmatrix} 1.49 & -0.74 & -0.66 & 0.16 & -0.98 & 0.65 & 0.15 & -0.08 \\ & 2.75 & 0.24 & -2.46 & 0.66 & -1.68 & -0.16 & 1.39 \\ & & 1.08 & 0.33 & 0.15 & -0.16 & -0.56 & -0.41 \\ & & & 2.6 & -0.08 & 1.39 & -0.41 & -1.53 \\ & & & & 2 & -0.82 & -1.18 & 0.25 \\ & \text{SYM} & & & & 3.82 & 0.33 & -3.53 \\ & & & & & & 1.59 & 0.25 \\ & & & & & & & 3.67 \end{bmatrix}.$$

- No body force, so $\mathbf{f}_\Omega = \mathbf{0}$. The only nonzero contribution to the boundary force comes from the traction on edge 1–4 (top), which in the parent domain corresponds to $\xi = -1$. Integrating analytically:

$$\mathbf{f}_\Gamma^e = \int_{\Gamma_{14}} (\mathbf{N}^{4Q})^T \bar{\mathbf{t}} \, d\Gamma = \int_{\eta=-1}^1 (\mathbf{N}^{4Q})^T (\xi = -1, \eta) \, d\eta \bar{\mathbf{t}} = \begin{bmatrix} 0 \\ -20 \\ 0 \\ 0 \\ 0 \\ 0 \\ -20 \end{bmatrix}.$$

Assembling and accounting for the reactions:

$$\mathbf{f}_\Gamma^e + \mathbf{r}^e = \begin{bmatrix} r_{x1} \\ r_{y1} - 20 \\ r_{x2} \\ r_{y2} \\ 0 \\ 0 \\ 0 \\ -20 \end{bmatrix}, \quad \mathbf{d} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ u_{x3} \\ u_{y3} \\ u_{x4} \\ u_{y4} \end{bmatrix}.$$

Example – Quadrilateral Element (cont.)

The global system of equations $10^7 \mathbf{K} \mathbf{d} = \mathbf{f} + \mathbf{r}$ is:

$$10^7 \begin{bmatrix} 1.49 & -0.74 & -0.66 & 0.16 & -0.98 & 0.65 & 0.15 & -0.08 \\ & 2.75 & 0.24 & -2.46 & 0.66 & -1.68 & -0.16 & 1.39 \\ & & 1.08 & 0.33 & 0.15 & -0.16 & -0.56 & -0.41 \\ & & & 2.6 & -0.08 & 1.39 & -0.41 & -1.53 \\ & & & & 2 & -0.82 & -1.18 & 0.25 \\ & \text{SYM} & & & & 3.82 & 0.33 & -3.53 \\ & & & & & & 1.59 & 0.25 \\ & & & & & & & 3.67 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ u_{x3} \\ u_{y3} \\ u_{x4} \\ u_{y4} \end{bmatrix} = \begin{bmatrix} r_{x1} \\ r_{y1} - 20 \\ r_{x2} \\ r_{y2} \\ 0 \\ 0 \\ 0 \\ -20 \end{bmatrix}.$$

The **reduced system** (F-nodes only):

$$10^7 \begin{bmatrix} 2 & -0.82 & -1.18 & 0.25 \\ & 3.82 & 0.33 & -3.53 \\ & & 1.59 & 0.25 \\ \text{SYM} & & & 3.67 \end{bmatrix} \begin{bmatrix} u_{x3} \\ u_{y3} \\ u_{x4} \\ u_{y4} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ -20 \end{bmatrix},$$

which yields

$$\begin{bmatrix} u_{x3} \\ u_{y3} \\ u_{x4} \\ u_{y4} \end{bmatrix} = 10^{-6} \begin{bmatrix} -1.17 \\ -9.67 \\ 2.67 \\ -9.94 \end{bmatrix}, \quad \mathbf{d}^e = 10^{-6} \begin{bmatrix} 0 \\ 0 \\ 0 \\ -1.17 \\ -9.67 \\ 2.67 \\ -9.94 \end{bmatrix}.$$

The resulting strains and stresses at the four Gauss points are

$$\boldsymbol{\varepsilon}^e(\xi_i, \eta_j) = \begin{bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \gamma_{xy} \end{bmatrix}_{(\xi_i, \eta_j)}^e = \mathbf{B}^e(\xi_i, \eta_j) \mathbf{d}^e, \quad \boldsymbol{\sigma}^e(\xi_i, \eta_j) = \begin{bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{xy} \end{bmatrix}_{(\xi_i, \eta_j)}^e = \mathbf{D}^e \boldsymbol{\varepsilon}^e(\xi_i, \eta_j).$$

$$\boldsymbol{\varepsilon}^e(\xi_1, \eta_1) = \mathbf{B}^e(\xi_1, \eta_1) \mathbf{d}^e = 10^7 \begin{bmatrix} -3.61 \\ -0.628 \\ -39.4 \end{bmatrix},$$

$$\boldsymbol{\sigma}^e(\xi_1, \eta_1) = \mathbf{D}^e \boldsymbol{\varepsilon}^e(\xi_1, \eta_1) = \begin{bmatrix} -12.5 \\ -5.64 \\ -45.5 \end{bmatrix},$$

$$\boldsymbol{\varepsilon}^e(\xi_1, \eta_2) = \mathbf{B}^e(\xi_1, \eta_2) \mathbf{d}^e = 10^7 \begin{bmatrix} 8.82 \\ -0.628 \\ -40.3 \end{bmatrix},$$

$$\boldsymbol{\sigma}^e(\xi_1, \eta_2) = \mathbf{D}^e \boldsymbol{\varepsilon}^e(\xi_1, \eta_2) = \begin{bmatrix} 28.5 \\ 6.65 \\ -46.5 \end{bmatrix}.$$

$$\boldsymbol{\varepsilon}^e(\xi_2, \eta_1) = \mathbf{B}^e(\xi_2, \eta_1) \mathbf{d}^e = 10^7 \begin{bmatrix} -11.7 \\ -3.45 \\ 2.21 \end{bmatrix},$$

$$\boldsymbol{\sigma}^e(\xi_2, \eta_1) = \mathbf{D}^e \boldsymbol{\varepsilon}^e(\xi_2, \eta_1) = \begin{bmatrix} -42.0 \\ -23.0 \\ 2.55 \end{bmatrix},$$

$$\boldsymbol{\varepsilon}^e(\xi_2, \eta_2) = \mathbf{B}^e(\xi_2, \eta_2) \mathbf{d}^e = 10^7 \begin{bmatrix} 6.65 \\ -3.46 \\ 0.95 \end{bmatrix},$$

$$\boldsymbol{\sigma}^e(\xi_2, \eta_2) = \mathbf{D}^e \boldsymbol{\varepsilon}^e(\xi_2, \eta_2) = \begin{bmatrix} 18.5 \\ -4.82 \\ 1.09 \end{bmatrix}.$$

Next: Finite Element Method for Linear Dynamics

Thanks for your attention!